

AUTONOMIC PHARMACOLOGY

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2.1 ORGANISATION & FUNCTION OF THE AUTONOMIC NERVOUS SYSTEM

LEARNING SCHEDULE - 16

AUTONOMIC NERVOUS SYSTEM ORGANISATION & FUNCTION

INTRODUCTION:

The autonomic nervous system is the subdivision of the peripheral nervous system that regulates body activities that are generally not under conscious control:

- Involuntary
- Functions to maintain homeostasis

Central control of the autonomic nervous system comes mostly from the hypothalamus, with some input from the limbic system and the reticular activating system

HOMEOSTASIS

The body's ability to maintain a stable, relatively constant condition through multiple dynamic equilibrium adjustment and regulation mechanisms.

NEUROTRANSMITTER

Neurotransmitters endogenous chemicals that transmit signals from a neuron to a target cell (FREQUENTLY another neuron or muscle) across a synapse.

- Examples Acetylcholine, epinephrine, norepinephrine, serotonin, dopamine etc.

Autonomic Nervous System (ANS)

- The ANS consists of motor neurons that:
 - Innervate smooth and cardiac muscle and glands
 - Make adjustments to ensure optimal support for body activities
 - Operate via subconscious control
 - Have viscera as most of their effectors

Divisions of the ANS

- **Sympathetic division** (thoracolumbar, “fight or flight”)
 - Thoracic and lumbar segments
- **Parasympathetic division** (craniosacral, “rest and repose”)
 - Preganglionic fibers leaving the brain and sacral segments
- **Enteric nervous system (ENS)**
 - May work independently

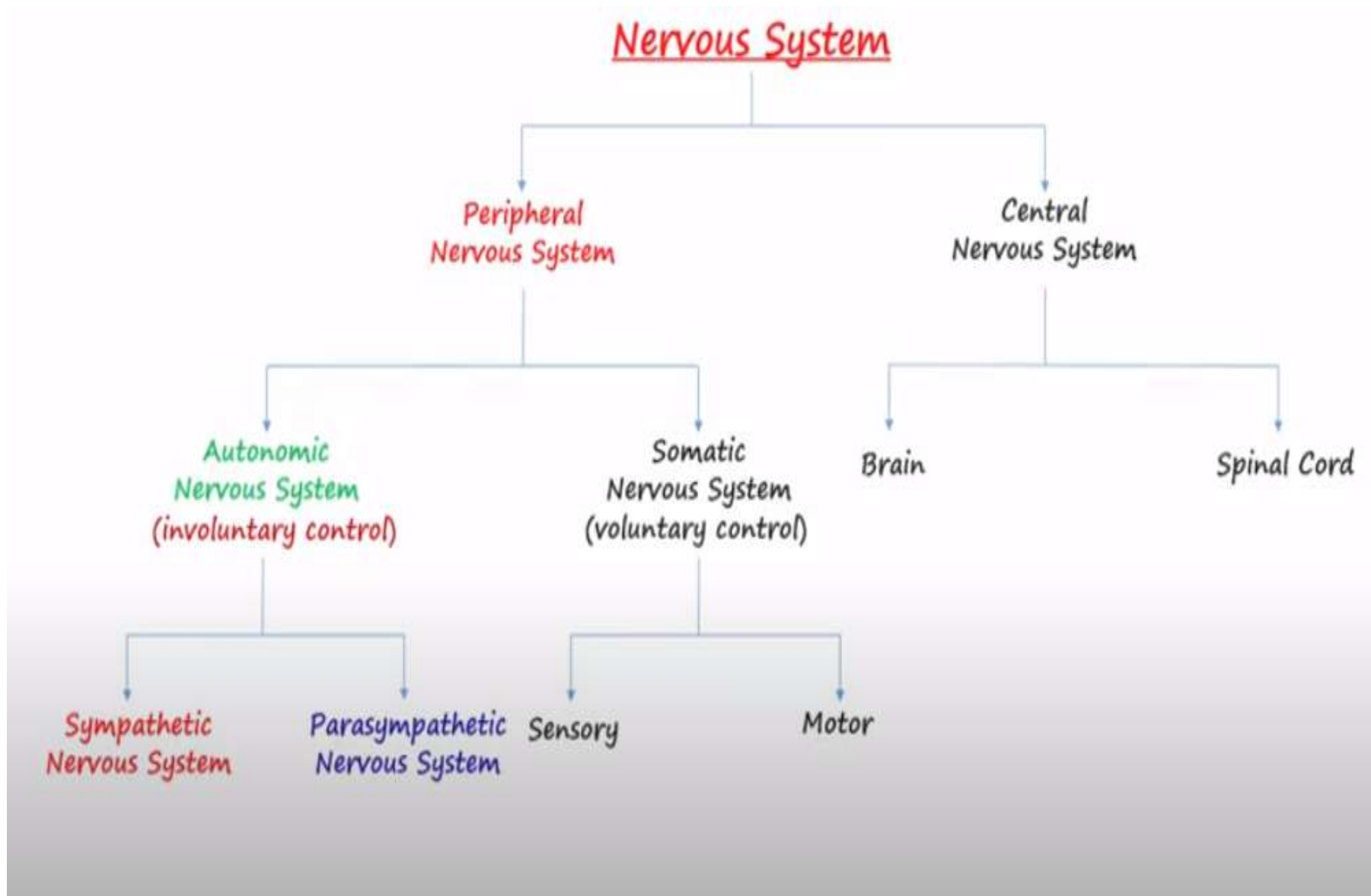
- Ganglia mass of neuron cell bodies that interconnect and relay info via synapses
- Dorsal Root ganglia afferent (sensory)
- Autonomic ganglia efferent (motor)
- Basal ganglion Brain-relay stations for multiple cerebral/cerebellar/thalamic/hypothalamic/brain stem functions
- Examples striatum, caudate nucleus, globus pallidum, substantia nigra, etc.
- Plexus intersecting and recombining nerve fibers- (major highway intersections)

AUTONOMIC NERVOUS SYSTEM ORGANISATION & FUNCTION

LEARNING OBJECTIVES:

- To describe the organization and function of the autonomic nervous system
- To describe the processes of chemical transmission in peripheral nervous system
- To describe how peripheral chemical transmission can be pharmacologically manipulated

Division of the Nervous System



ANATOMICAL DISTRIBUTION OF THE AUTONOMIC NERVOUS SYSTEM

Parasympathetic neurons (PSNS)

Originate in the midbrain, medulla oblongata and sacral spinal cord

Sympathetic neurons (SNS)

Originate from the thoracic and lumbar portions of the spinal cord

Sympathetic and Parasympathetic

- Often they have opposing effects
- May work independently
- May work together each one controlling one stage of the process

ANS Versus Somatic Nervous System (SNS)

- The ANS differs from the SNS in the following three areas
 - Effectors
 - Efferent pathways
 - Target organ responses

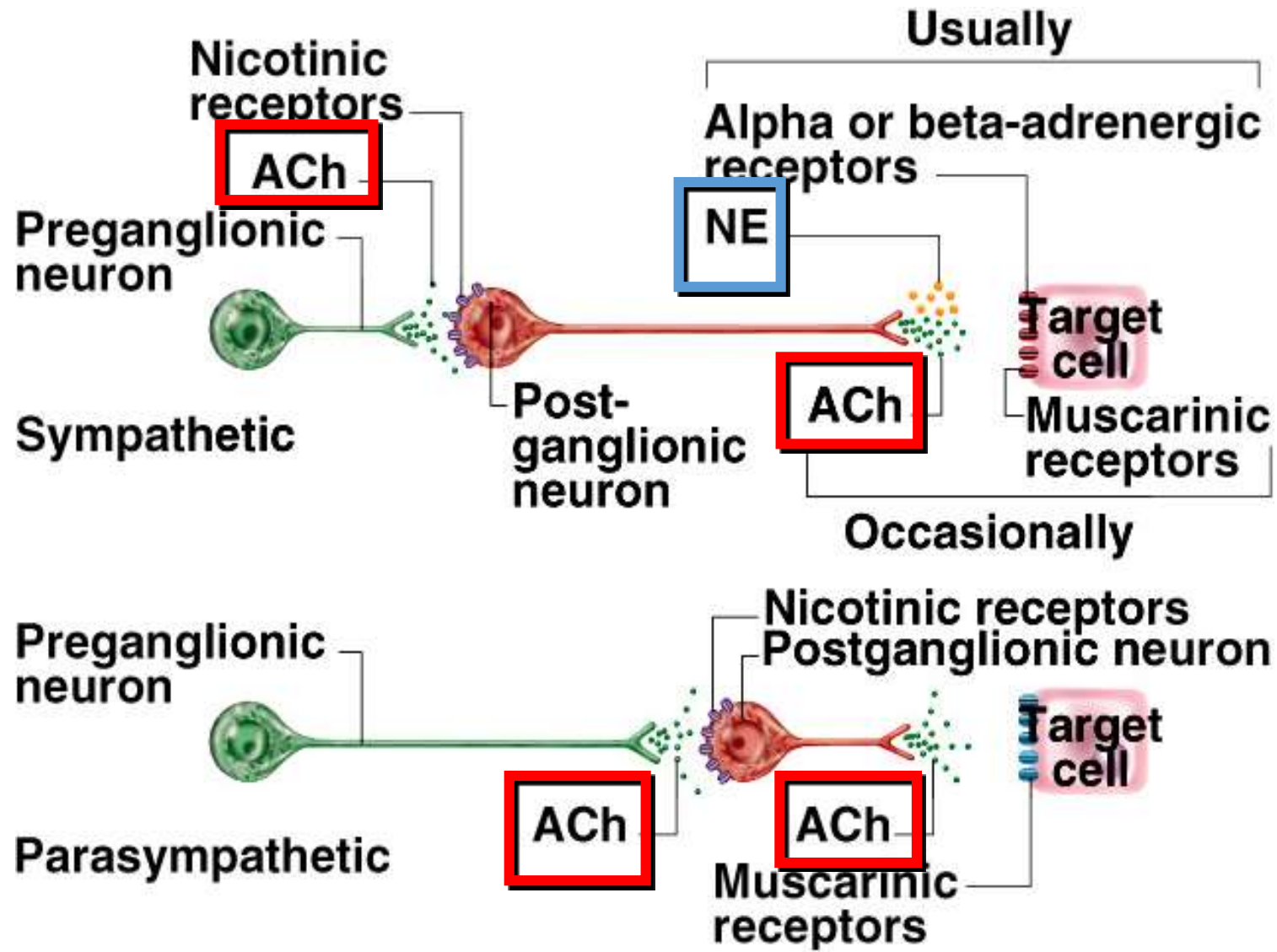
Effectors

- The effectors of the SNS are skeletal muscles
- The effectors of the ANS are cardiac muscle, smooth muscle, and glands

Efferent Pathways

- Heavily myelinated axons of the somatic motor neurons extend from the CNS to the effector
- Axons of the ANS are a two-neuron chain
 - The **preganglionic** (first) neuron has a lightly myelinated axon
 - The **ganglionic** (second) neuron extends to an effector organ

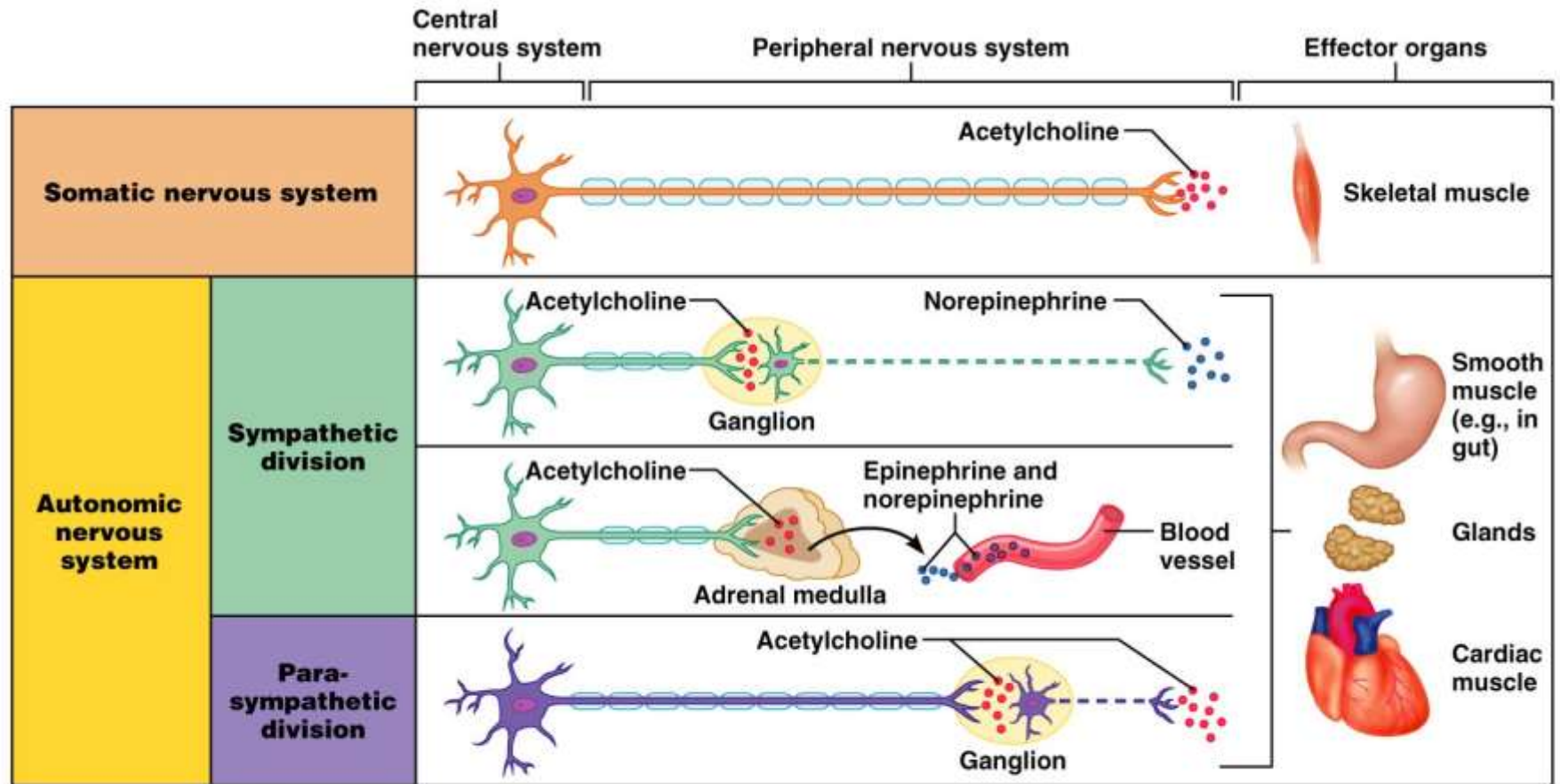
Neurotransmitters and Receptors



Neurotransmitter Effects

- All somatic motor neurons release Acetylcholine (ACh), which has an excitatory effect
- In the ANS:
 - Preganglionic fibers release ACh
 - Postganglionic fibers release norepinephrine or ACh and the effect is either stimulatory or inhibitory
 - ANS effect on the target organ is dependent upon the neurotransmitter released and the receptor type of the effector

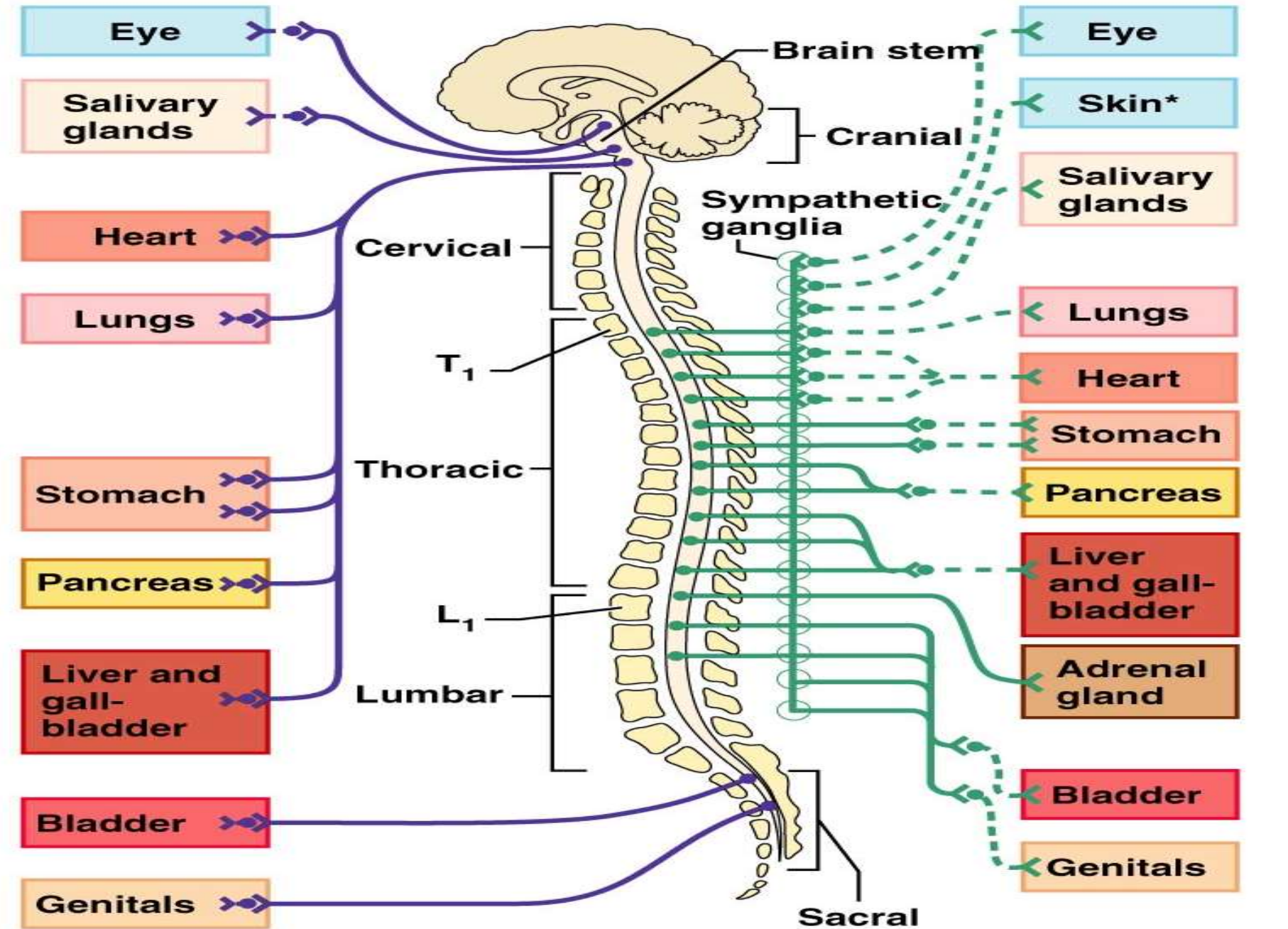
Comparison of Somatic and Autonomic Systems



Key:

— = Preganglionic axons (sympathetic)
 --- = Postganglionic axons (sympathetic)
  = Myelination
 — = Preganglionic axons (parasympathetic)
 --- = Postganglionic axons (parasympathetic)

Parasympathetic



- In general, **PSNS** and **SNS** nerves control the same organs, but have opposing effects on them!
 - SNS is fight or flight
 - PSNS is rest and digest, feed and breed
- All nerve pathways originate from the CNS (brain + spinal cord)
 - 12 cranial
 - 0 cervical
 - 12 thoracic
 - 5 lumbar
 - 5 sacral



ANATOMICAL DISTRIBUTION OF THE AUTONOMIC NERVOUS SYSTEM ... CONT'D

Autonomic system consists of chains of two motor neurons:

1st: Preganglionic neuron (in brain or spinal cord)

2nd: Postganglionic neuron (cell body in ganglion outside CNS)

- Axon of 1st (preganglionic) neuron leaves CNS to synapse with the 2nd (postganglionic) neuron
- Axon of 2nd (postganglionic) neuron extends to the organ it serves

FUNCTIONS OF THE PARASYMPATHETIC NERVOUS SYSTEM

REST and DIGEST

- Eye: constriction of the pupil (miosis) and accommodation
- Lacrimation
- Salivation
- Slowing of the heart (bradycardia)
- Decrease in blood pressure
- Production of bronchial secretions

FUNCTIONS OF THE PARASYMPATHETIC NERVOUS SYSTEM ... CONT'D

- Production of digestive juices from stomach, pancreas and gallbladder
- Increased gastrointestinal tract tone and peristalsis
- Sphincter relaxation
- Urination and defecation
- Penile erection

FUNCTIONS OF THE SYMPATHETIC NERVOUS SYSTEM

Helps the body cope with external stimuli and functions during stress (triggers the flight or fight response)

- Increased heart rate (tachycardia)
- Vasoconstriction of skin and viscera
- Vasodilatation in skeletal muscle
- Sweating

FUNCTIONS OF THE SYMPATHETIC NERVOUS SYSTEM ... CONT'D

- Bronchodilatation
- Dilation of the pupil (mydriasis)
- Inhibition of production of digestive juices
- Inhibition of peristalsis
- Relaxation of the uterus

INTEGRATION OF AUTONOMIC FUNCTION

Generally the parasympathetic system causes discrete changes in organ function and usually associated with “rest and digest”:

- Increased salivary secretion
- Increased GI motility
- Increased urination, defecation
- Increased bronchial constriction, secretions
- Decreased heart rate
- Pupillary constriction, decreased visual accommodation
- Increased sexual arousal

INTEGRATION OF AUTONOMIC FUNCTION CONT'D

Parasympathetic activity does not usually cause generalized organ system activation unless:

- Acetylcholine metabolism is significantly inhibited (nerve gas, insecticides)
- Exposure to muscarinic agonists (muscarine from mushrooms)

INTEGRATION OF AUTONOMIC FUNCTION

.... CONT'D

Sympathetic activation prepares the organism to flee from or fight a potential threat:

- Pupils dilate, lens flatten for far-vision
- Heart rate, force increased
- Vasoconstriction in most vascular beds but vasodilatation in skeletal muscle
- Blood shunted from viscera to skeletal muscle
- Increased respiration rate
- Glucose release by liver

INTEGRATION OF AUTONOMIC FUNCTION

.... CONT'D

- Kidney blood flow and water loss decreased
- GI and bladder function decreased
- Skin blood flow decreased, pilo-erection

Sympathetic nervous system is designed to provide more wide-spread activation of organ systems during severe stress

This is assured by interconnections between sympathetic ganglia in the chain ganglia that allow almost simultaneous activation of multiple nerve bundles supplying different organs

LEARNING OUTCOMES

Students must be able to comprehend:

- The organization and function of the autonomic nervous system
- The processes of chemical transmission in peripheral nervous system
- How peripheral chemical transmission can be pharmacologically manipulated

END

Thanks for listening

